

CRANBROOK SCHOOL

Year 12 MATHEMATICS - 2 Unit/3 Unit Common Trial

Term 3 1999

Time: 3 h /SKB,HRK,MJB,GJB,KMR,WMF.

All questions are of equal value.

All necessary working should be shown in every question.

Full marks may not be awarded if work is careless or badly arranged.

Approved silent calculators may be used.

Submit your work in six bundles .

Q 1-3 : 8 page booklet

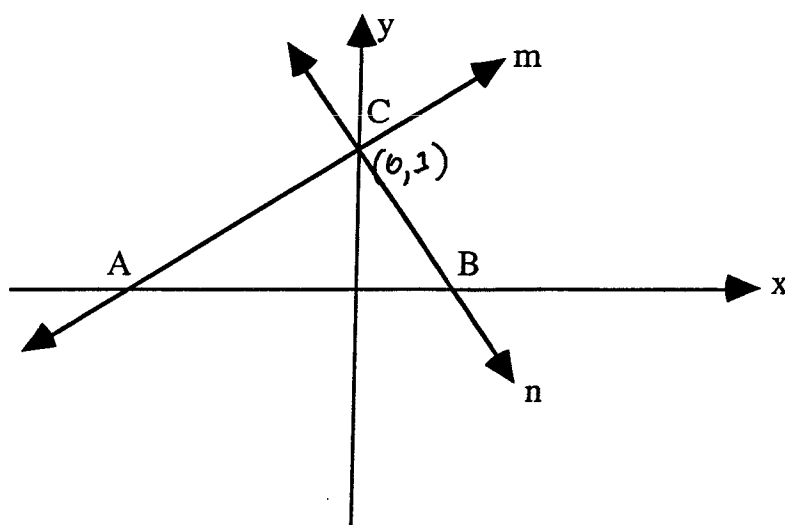
Q 4-6 : 8 page booklet

Q 7; Q 8; Q 9 ; Q 10. : each a 4 page booklet.

[Use an 8 page booklet for Questions 1 - 3]

1. (a) What is the value of π^2 correct to 2 significant figures ? [2 marks]
- (b) Simplify $(10 - 2y) - (8y - 4)$ [2 marks]
- (c) A six-sided die is thrown twice. What is the probability that the same face is uppermost each time ? [2 marks]
- (d) Find the exact value of 225° in radians ? [1 mark]
- (e) Find the primitive of $\sin 3x$. [2 marks]
- (f) If $3^x = \frac{1}{3}$ find the value of x . [1 mark]
- (g) A Holden's value depreciates by 15 % p.a.
If it was worth \$10 000 in 1998 what will it be worth in 2000 ? [2 marks]
2. (a) Differentiate the following functions with respect to x simplifying your answers where applicable :
- (i) $y = (3x^5 - 1)^7$ (ii) $y = \sin^2 5x$ (iii) $y = \frac{3}{\cos x}$ [6 marks]
- (b) Find $\int \tan^2 x \, dx$ [2 marks]
- (c) Evaluate $\int_0^1 e^{\frac{1}{2}x} \, dx$ [2 marks]
- (d) Find $\int \frac{10x^2 - 1}{10x^3 - 3x} \, dx$ [2 marks]

3. On the diagram shown below the lines m and n meet at the point C.
The co-ordinates of C are $(0, 3)$.



- (a) If the gradient of line m is $\frac{3}{4}$ what is the equation of line m ? [2 marks]
- (b) If line m is perpendicular to line n what is the equation of line n ? [2 marks]
- (c) What are the co-ordinates of A? [1 mark]
- (d) What are the co-ordinates of B? [1 mark]
- (e) What is the area of triangle ABC? [2 marks]
- (f) Find the perpendicular distance from A to line n . [2 marks]
- (g) Write down a pair of inequalities which define the region between lines m and n which also includes the origin. [2 marks]

[Use a new 8 page booklet for Questions 4 - 6]

4. (a) If α and β are the roots of the equation $5x^2 - 7x + 4 = 0$,
find the value of: $\alpha^2 + \beta^2$ [3 marks]
- (b) The gradient of a curve is given by $\frac{dy}{dx} = 5x - 4$.
Find the equation of the curve if it passes through $(-2, 5)$. [2 marks]
- (c) The parabola S has equation $4x^2 = y + 2$.
- (i) What are the co-ordinates of the vertex of S? [1 mark]
- (ii) What are the co-ordinates of the focus of S? [2 marks]
- (d) Find the equation of the tangent to the curve $y = \log_e 2x$ at the point where $x = 3$.
Write this equation in general form. [4 marks]

5. (a) If the area under the curve $y = x^2 - 9$ from $x = 1$ to $x = 3$ is rotated about the y-axis, calculate the volume of the solid of revolution formed. [3 marks]
- (b) (i) Find an expression in terms of a , r , and n , for the difference between the Limiting Sum and the sum to n terms of a G. P. [2 marks]
- (ii) Hence or otherwise find the difference between the limiting sum and the sum of the first five terms of the sequence $\frac{2}{3}, \frac{1}{2}, \frac{3}{8}, \dots$ [3 marks]
- (c) Use Simpson's Rule with five function values to find an approximate value for π using the formula below. (Give your answer correct to 5 significant figures.)

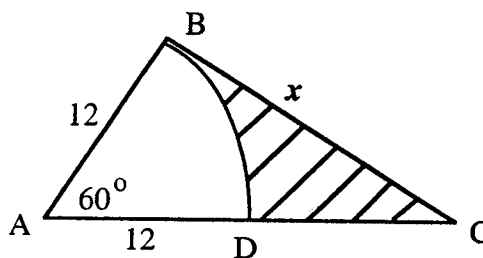
$$\pi = 4 \int_0^1 \frac{dx}{x^2 + 1} \quad [4 \text{ marks}]$$

6. (a) Solve $\log_e(x^2 - x) = \log_e 2 + \log_e(3x + 4)$ [2 marks]
- (b) A can is to be made in the form of a cylinder of volume 512 cm^3 . The can is to have no top and the radius, r , of the base is restricted to $2 \leq r \leq 8$.
- (i) If S represents the area of metal required to make the cylinder, show that
$$S = \pi r^2 + \frac{1024}{r}$$
 [3 marks]
- (ii) The formula in (i) has one turning point in the given Domain. Find the coordinates of this turning point. [3 marks]
- (iii) Sketch the graph of the area, S , against the radius, r , in the given Domain, showing clearly the turning point and the end points. [3 marks]
- (iv) Find the value of r for which the area of metal is greatest. [1 mark]

[Use a new 4 page booklet for Question 7]

7. (a) In the diagram ABD is a sector of a circle of a circle, centre A.

$AB = AD = 12 \text{ cm}$
 $\hat{BAC} = 60^\circ, \hat{ABC} = 90^\circ$
 $BC = x \text{ cm}$



- (i) Copy the diagram into your examination book and then calculate the exact value of x . [2 marks]
- (ii) Convert angle A to radian measure and hence calculate the shaded area. [4 marks]
- (b) If $f(x) = ax^2 + bx + c$ and $a > 0$, show that the minimum value of the function is $\frac{4ac - b^2}{4a}$. [3 marks]
- (c) For what values of k are the roots of $(k + 1)x^2 - 4kx + 3k = 0$ real and different? [3 marks]

[Use a new 4 page booklet for Question 8]

8. (a) Two straight roads AB and AC are inclined to each other at $62^\circ 35'$. Two bike riders start simultaneously from A and travel along the roads at 15 km/h and 25 km/h respectively.

How long is it before they are 90 km apart?
(Give your answer correct to the nearest minute.)

[5 marks]

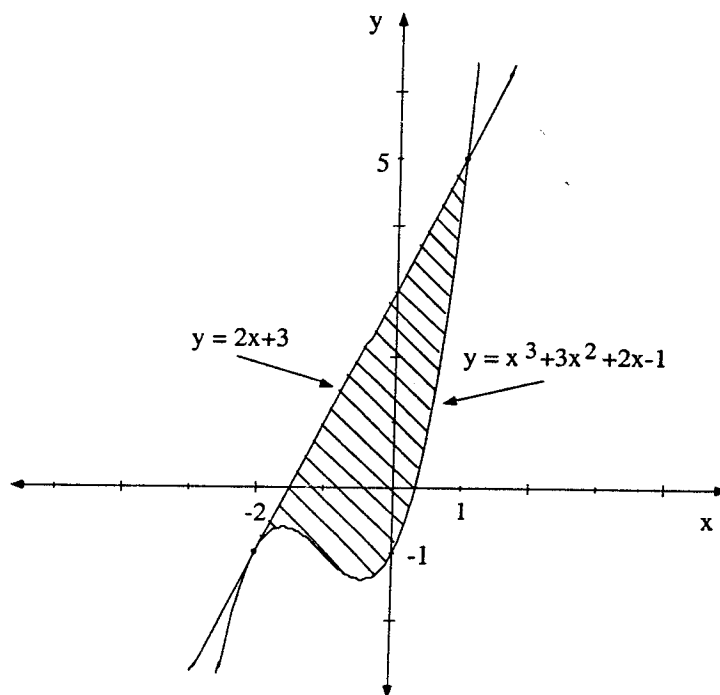
- (b) Show that the curve $y = e^{-x^2}$ has one maximum turning point and two points of inflexion. Find the co-ordinates of these points and sketch the curve.

[7 marks]

[Use a new 4 page booklet for Question 9]

9. (a) As shown in the diagram below, $y = 2x + 3$ is a tangent to the curve $y = x^3 + 3x^2 + 2x - 1$ at the point $(-2, -1)$ and cuts the curve again at $(1, 5)$.

Find the area enclosed by these two curves.



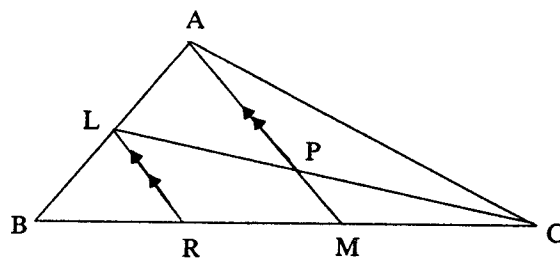
[4 marks]

- (b) In the diagram below,

$LR \parallel AM$,

$BM : MC = 10 : 9$ and

$BL : LA = 3 : 2$.



- (i) Prove that $\triangle BLR$ is similar to $\triangle BAM$. [2 marks]
- (ii) What is the ratio of $BR : RM$. State the geometric fact(s) used. [2 marks]
- (ii) Stating all geometric facts used, find the ratio of $MP : RL$. [4 marks]

[Use a new 4 page booklet for Question 10]

10. (a) A crab fisherman nets 25 crabs of which 5 are female which, at this time of year, should be returned to the water. His mate takes 15 crabs, 4 of them female, and both bring their catches ashore in hessian sacks. The fishing inspector is awaiting them, tosses a coin to decide whose catch he will examine, and then randomly pulls a crab out of the sack. If it is a female he will fine the fisherman and also take one crab out of the other's sack to check it as well. By considering the possible outcomes, or otherwise,

- (i) What is the probability that neither of the fishermen is fined?
- (ii) What is the probability that both of the fishermen are fined? [6 marks]

- (b) A couple takes a mortgage of \$150 000 to buy a house.

The loan is to be repaid in equal weekly installments over a 15 year period at an annual rate of interest of 6.24%.

Using 1 year = 52 weeks, it is calculated that the weekly repayments would be \$296.25 over the 15 year period. However the couple decides to pay weekly installments of \$320.

- (i) How long will it take them to repay the loan?
- (ii) How much will they save compared with the 15 year period? [6 marks]

1. (a) $\pi^2 = 9.869604401 \dots$
 $= 9.9$ (2 sig. figs) (2m)

(b) $(10-2y) - (8y-4)$
 $= 14 - 10y$ (2m)

(c) $P(\text{same face}) = 6 \times \frac{1}{6} \times \frac{1}{6}$
 $= \frac{1}{6}$ (2m)

(d) $225^\circ = 225^\circ \times \frac{\pi}{180^\circ}$
 $= \frac{5\pi}{4}$ (1m)

(e) let $\frac{dy}{dx} = \sin 3x$
 $\therefore y = -\frac{1}{3} \cos 3x + c$ (2m)

(f) $3^x = \frac{1}{3}$
 $\therefore 3^x = 3^{-1}$
 $\therefore x = -1$ (1m)

(g) $D = P(1 - \frac{r}{100})^n$
 $\therefore D = 10000(1 - \frac{15}{100})^2$
 $= 7225$

$\therefore$ Holden is worth \$7225 in 2000. (2m)

2. (a) (i) let $y = (3x^5 - 1)^7$
 $\therefore \frac{dy}{dx} = 7(3x^5 - 1)^6 \cdot 15x^4$
 $= 105x^4(3x^5 - 1)^6$ (2m)

(ii) let $y = \sin^2 5x$
 $\therefore y = (\sin 5x)^2$
 $\therefore \frac{dy}{dx} = 2 \sin 5x \cdot 5 \cos 5x$
 $= 10 \sin 5x \cos 5x$ (2m)

(iii) let $y = \frac{3}{\cos x}$
 $\therefore y = 3 \sec x$

$\therefore \frac{dy}{dx} = 3 \sec x \tan x$ (2m)

(b) $\int \tan^2 x \, dx$
 $= \int \sec^2 x - 1 \, dx$
 $= \tan x - x + c$ (2m)

(c) $\int_0^1 e^{\frac{1}{2}x} \, dx$
 $= [2e^{\frac{1}{2}x}]_0^1$
 $= [2e^{\frac{1}{2}} - 2e^0]$
 $= 2[e^{\frac{1}{2}} - 1]$ (2m)

(d) $\int \frac{10x^2 - 1}{10x^3 - 3x} \, dx$
 $= \frac{1}{3} \int \frac{30x^2 - 3}{10x^3 - 3x} \, dx$
 $= \frac{1}{3} \ln |10x^3 - 3x| + c$ (2m)

3. (i) eq'n of line m is:
 $y - 3 = \frac{3}{4}(x - 0)$
 $\therefore 4y - 12 = 3x$
 $\therefore 3x - 4y + 12 = 0$ (2m)

(ii) gradient of line n is $-\frac{4}{3}$
 $\therefore$ eq'n of line n is:
 $y - 3 = -\frac{4}{3}(x - 0)$
 $\therefore 3y - 9 = -4x$
 $\therefore 4x + 3y - 9 = 0$ (2m)

(iii) As m meets x-axis at
sub $y=0$ into m's eq'n.

$$\therefore 3x + 12 = 0$$

$$\therefore x = -4$$

$$\Rightarrow A \text{ is } (-4, 0). \quad (1m)$$

(iv) Similarly sub. $y=0$ into
n's eq'n.

$$\therefore 4x - 9 = 0$$

$$\therefore x = +2\frac{1}{4}$$

$$\Rightarrow B \text{ is } (2\frac{1}{4}, 0). \quad (1m)$$

$$\begin{aligned} \text{(v) Area of } \triangle ABC &= \frac{1}{2}bh \\ &= \frac{1}{2} \times 6\frac{1}{4} \times 3 \\ &= 9\frac{3}{8} \text{ units}^2 \\ &\quad (2m) \end{aligned}$$

(vi) As $m \perp n$ the perp. distance
from A to line n is the same
as the distance between A and C.

$$\begin{aligned} \therefore d &= \sqrt{(-4-0)^2 + (0-3)^2} \\ &= \sqrt{16+9} \\ &= 5 \end{aligned}$$

$$\therefore \text{perp. distance is } 5 \text{ units.} \quad (2m)$$

(vii) Test $(0,0)$ in $3x-4y+12 > 0$

$$\therefore 12 > 0 \text{ true}$$

$\therefore$ m's inequality is $3x-4y+12 > 0$

Test $(0,0)$ in $4x+3y-9 < 0$

$$\therefore -9 < 0 \text{ true}$$

$\therefore$ n's inequality is $4x+3y-9 < 0$

4.2
 $\alpha + \beta = \frac{7}{5}$

$\alpha\beta = \frac{4}{5}$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= \frac{9}{25}$

b) $y = \frac{5x^2}{2} - 4x + c$

$\left. \begin{matrix} x=2 \\ y=5 \end{matrix} \right\} C = -13$

c) $x^2 = \frac{1}{4}(y+2)$

Vertex $(0, -2)$

$a = \frac{1}{16}$

Focus $(0, -1\frac{15}{16})$

d) $y = \log_e 2x$

$= 3 \} y = \log_e 6$

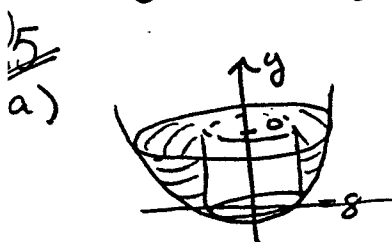
$\frac{dy}{dx} = \frac{1}{2x} \times 2$

$= \frac{1}{x}$

$\therefore x=3, m = \frac{1}{3}$

$y - \log_e 6 = \frac{1}{3}(x-3)$

$x - 3y - 3 + 3\log_e 6 = 0$



$V = \pi \int_{-8}^0 (y+9 - \pi \cdot 1.8) dy$

$= \pi \left(\frac{y^2}{2} + 9y \right)_{-8}^0 - 8\pi$

$= 40\pi - 8\pi$

$= 32\pi u^3$

b)

$S_{\infty} - S_n$

$= \frac{a}{1-r} - \frac{a-ar^n}{1-r}$

$= \frac{ar^n}{1-r}$

(ii) $a = \frac{2}{3} \quad r = \frac{3}{4}$

$S_{\infty} - S_n = \frac{81}{128}$

c) $\pi \approx 4 \int_0^1 \frac{dx}{x^2+1}$

$= 4 \left(\frac{1}{3} \right) \left[1 + \frac{1}{2} + 4 \left(\frac{16}{17} + \frac{16}{25} \right) + 2 \left(\frac{4}{5} \right) \right]$

$= 3.1416 \text{ (5 sig figs)}$

6(a) $\log(x^2 - x) = \log 2(3x+4)$

$x^2 - x = 6x + 8$

$x^2 - 7x - 8 = 0$

$(x-8)(x+1) = 0$

$x = 8, -1$

b) $V = 512 = \pi r^2 h$

$h = \frac{512}{\pi r^2}$

Area of base $= \pi r^2$

Curved area $= 2\pi r h$

$= 2\pi r \frac{512}{\pi r^2}$

$= \frac{1024}{r}$

$\therefore \text{TSA} = \pi r^2 + \frac{1024}{r}$

(ii)

$\frac{ds}{dr} = 2\pi r - \frac{1024}{r^2}$

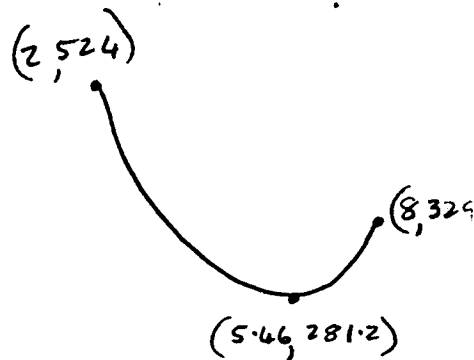
$2\pi r - \frac{1024}{r^2} = 0$

$r = \sqrt[3]{\frac{512}{\pi}} = \frac{8}{\sqrt{\pi}} \approx 5.4$

Then $S \approx 281.2$

When $r = 2 \quad S \approx 524$

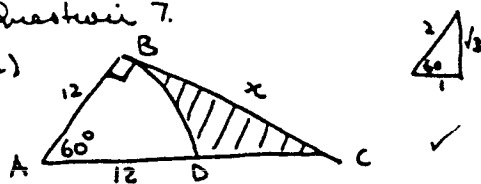
$r = 8 \quad S \approx 329$



(iv) $r = 2$

Question 7.

(a)



(i) $\tan 60 = \frac{x}{12}$

$x = 12 \tan 60$
 $= 12\sqrt{3} \text{ cm}$ ✓

non exact

units

(ii) $180^\circ = \pi \text{ c}$
 $60^\circ = \frac{\pi \text{ c}}{3}$ ✓

1.047

Area $\Delta = \frac{12 \cdot 12\sqrt{3}}{2}$ ✓
 $= 72\sqrt{3}$ 124.7

Area sector = $\frac{1}{2} r^2 \cdot \frac{\pi \text{ c}}{3}$ ✓
 $= 24\pi$ 75.4

Shaded area = $72\sqrt{3} - 24\pi$ ✓
 $= 24(3\sqrt{3} - \pi) \text{ cm}^2$ 49.3

(b) $f(x) = ax^2 + bx + c$

Quadratic $\Rightarrow$ axis of symmetry

$x = -\frac{b}{2a}$ ✓

Min value $y = a\left(-\frac{b}{2a}\right)^2 + b\left(-\frac{b}{2a}\right) + c$ ✓

so solve for T.P.

$= \frac{+2ab^2}{4a^2} - \frac{b^2}{2a} + c$

$= \frac{+b^2 - 2b^2 + 4ac}{4a}$

$= \frac{4ac - b^2}{4a}$ ✓

(c) Real and different

$b^2 > 4ac$

$(-4k)^2 =$

$\Rightarrow (-4k)^2 > 4(k+1)3k$ ✓

$16k^2 > 12k^2 + 12k$

$(+12)$

$4k^2 - 12k > 0$

$4k(k-3) > 0$ ✓

$0 > k$ or $k > 3$ ✓

inequality

(b) find $f'(x)$ then not find value of x .

8 (a) Let t hrs be the required time

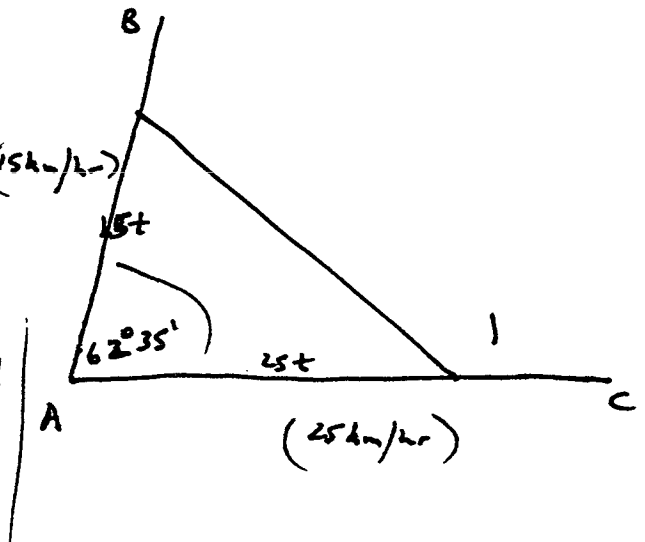
Dist. travelled along AB = $15t$ km

" " " AC = $25t$ km

If distance between = 90 km

$$8100 = (15t)^2 + (25t)^2 - 2(15t)(25t)\cos 62^\circ 35'$$

$$= t^2(225 + 625 - 750\cos 62^\circ 35')$$



$$\therefore t^2 = \frac{8100}{850 - 750\cos 62^\circ 35'}$$

$$= 16.0505$$

$$\therefore t = 4.006$$

is. Time = 4 hrs 0 mins (nearest minute)

(b)

$$y = e^{-x^2}$$

$$\frac{dy}{dx} = -2x e^{-x^2}$$

$$\frac{d^2y}{dx^2} = -2 \left[x \cdot (-2x) e^{-x^2} + e^{-x^2} \cdot 1 \right]$$

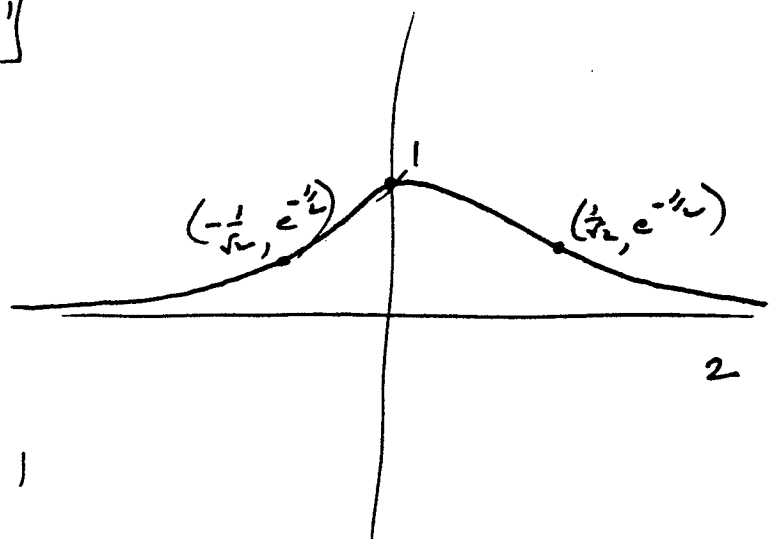
$$= -2 e^{-x^2} (1 - 2x^2)$$

Stat pt - one, viz $(0, 1)$

Since $\frac{d^2y}{dx^2} = -2 < 0$

$(0, 1)$ is a maximum

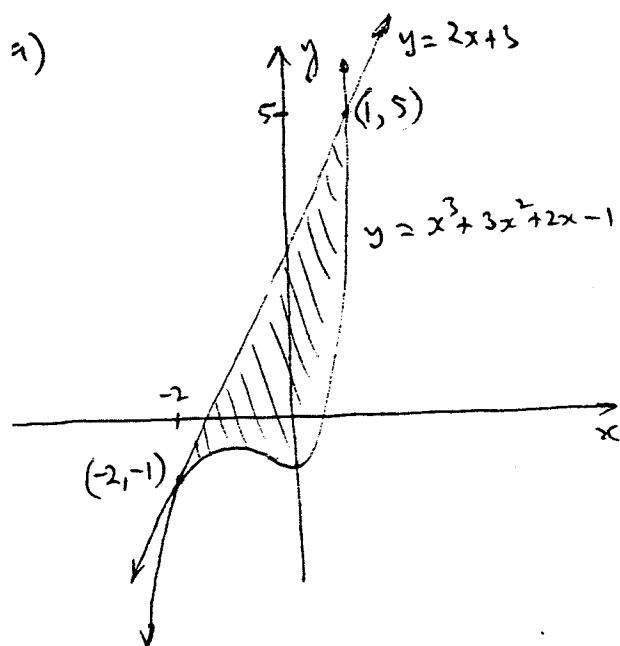
Inflexions - two, viz $\left(\frac{1}{\sqrt{2}}, e^{-1/2}\right)$
and $\left(-\frac{1}{\sqrt{2}}, e^{-1/2}\right)$



$$\begin{cases} x \rightarrow \pm\infty & y \rightarrow 0 \\ y > 0 & \forall x \end{cases}$$

1999 2U HSC TRIAL - QUESTION 9.

VI
2u



The Shaded area is given by

$$= \int_{-2}^1 [(2x+3) - (x^3+3x^2+2x-1)] dx$$

$$= \int_{-2}^1 (2x+3-x^3-3x^2-2x+1) dx$$

$$= \int_{-2}^1 (-x^3-3x^2+4) dx \quad [2]$$

$$= \left[-\frac{x^4}{4} - x^3 + 4x \right]_{-2}^1 \quad [1]$$

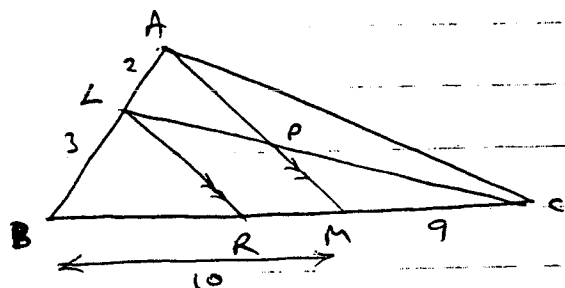
$$= \left(-\frac{1}{4} - 1 + 4 \right) - \left(-\frac{(-2)^4}{4} - (-2)^3 + 4(-2) \right)$$

$$= -\frac{1}{4} - 1 + 4 + 4 - 8 + 8$$

$$= 6\frac{3}{4}$$

∴ the area is $6\frac{3}{4}$ units² [1]

(b)



(i) RTP $\triangle BLR \parallel \triangle BAM$

Proof: In $\triangle BLR$ & $\triangle BAM$

$\angle LBR = \angle ABM$ (same angle)

$\angle BLR = \angle BAM$ } Corresponding angles are equal:
 $\angle BRL = \angle BMA$ } $LR \parallel AM$

∴ $\triangle BLR \parallel \triangle BAM$ (Squaring)

[2] (Deduct errors)

(ii) $BR:RM = BL:LA = 3:2$ [1]

(A line LR parallel to one side AM of a triangle cuts off intercepts on the other sides in the same ratio) [1]

(iii) $BR:RM = 3:2$

∴ $BR:BM = 3:5$ but $BM = 10$

$$\therefore \frac{BR}{10} = \frac{3}{5}$$

$$\therefore BR = 6$$

$$\therefore RM = 4$$

and so $CM:CA = 9:13$ [2]

Now in $\triangle CPM$ and $\triangle CLR$.

$\angle PCM = \angle CLR$ (same angle)

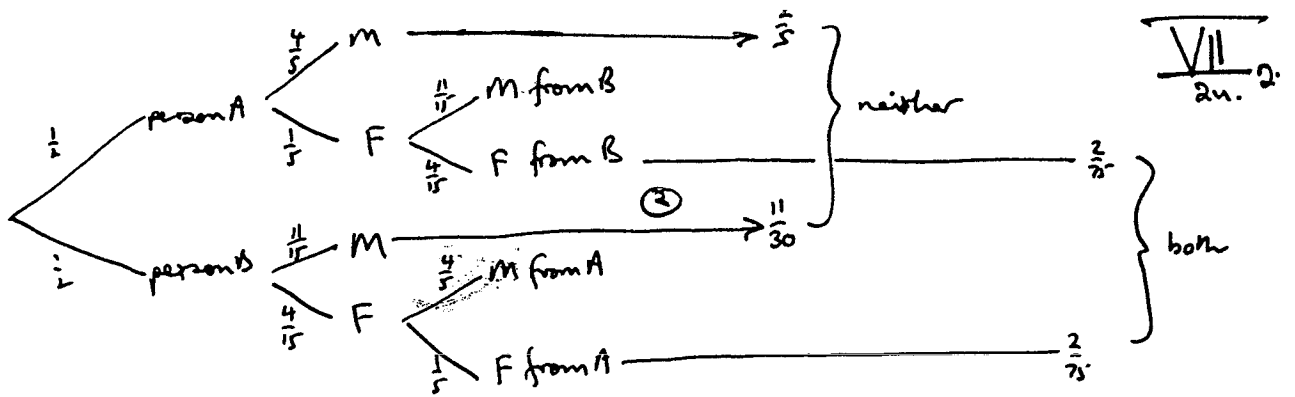
$\angle CPM = \angle CLR$ } Corresponding angles are equal:
 $\angle CMP = \angle CLR$ } $PM \parallel LR$

∴ $\triangle CPM \parallel \triangle CLR$ [1]

∴ $MP:RL = CM:MR$ (Corresponding sides of similar triangles are in the same ratio)

[1]

(a)



(i) Neither: $\frac{2}{5} + \frac{11}{20} = \frac{23}{20}$ ①

(ii) Both: $\frac{2}{25} + \frac{2}{25} = \frac{4}{25}$ ②

(b) Interest = $\frac{6.24}{52} = 0.12\%$ per week

Owing after 1 week = $150000(1.0012) - 320$ ①

" " 2 " = $(150000(1.0012) - 320)(1.0012) - 320$

" " n weeks = $150000(1.0012)^n - 320(1 + 1.0012 + \dots + 1.0012^{n-1})$ ①

= 0 if repaid

$150000(1.0012)^n = 320 S_n$ ① ——— ①

where S_n is sum geom series $a=1$
 $r=1.0012$

ie $\frac{1(1.0012^n - 1)}{0.0012}$ ①

① becomes $150000(1.0012)^n = \frac{320}{0.0012} (1.0012^n - 1)$

$0.5625 \times 1.0012^n = 1.0012^n - 1$

$(1.0012)^n (1 - 0.5625) = 1$

$1.0012^n = \frac{1}{0.4375}$

$n \log(1.0012) = \log\left(\frac{1}{0.4375}\right)$

$n = 689.31...$

= 690 weeks (nearest) ①

Total paid = $n \times 320 = \$220579.86$
 = $\$220580$ near enough

Original total = $15 \times 52 \times \$296.25 = 231075$

$\therefore$ Savings = $\$10495$ ①